

Resonance

EEE203

Resonance

- For a particular range of frequencies, the response will be near or equal to the maximum. The frequencies to the far left or right have very low voltage or current levels and, for all practical purposes, have little effect on the system's response.
- When resonance occurs due to the application of the proper frequency, the energy absorbed by one reactive element is the same as that released by another reactive element within the system.

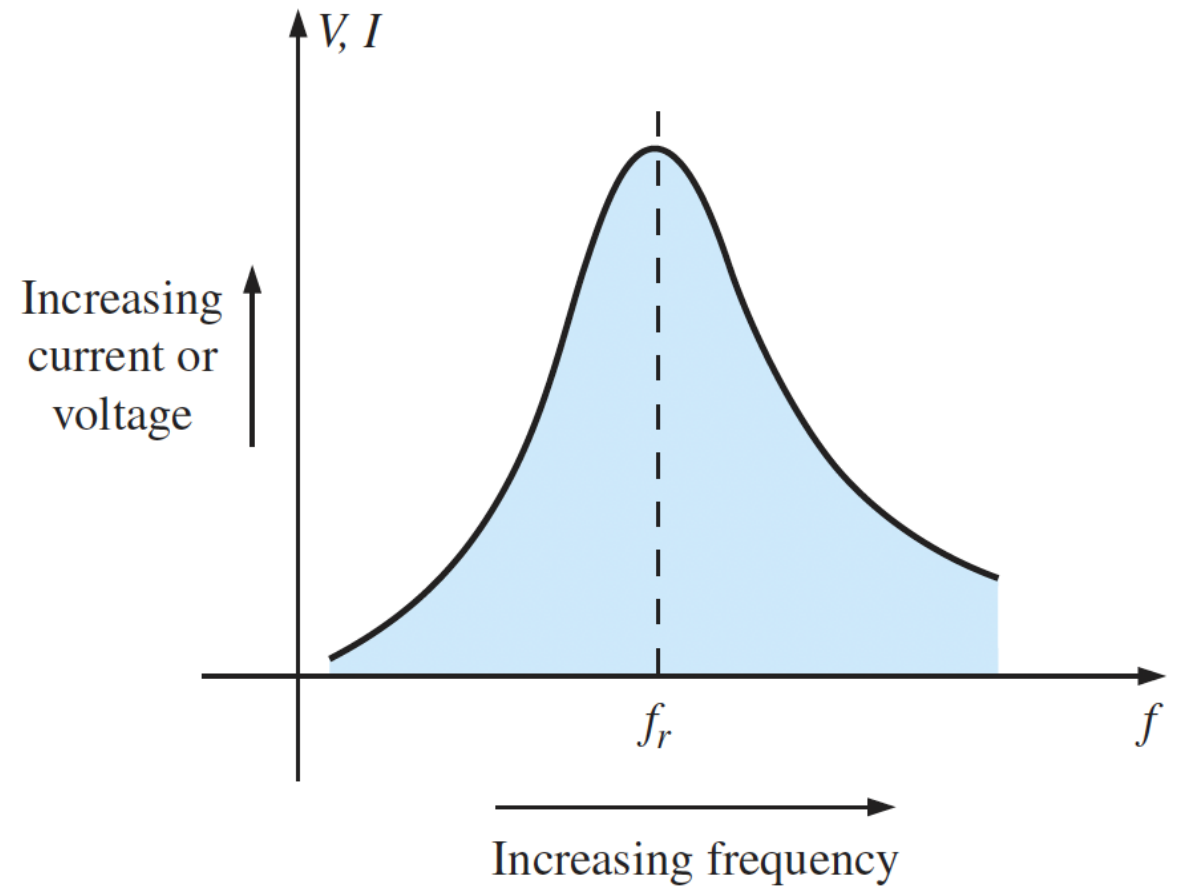


Fig. Resonance curve

Resonance

- There are **two types** of resonant circuits: **series and parallel**.
- A series resonant circuit is one where the resonant curve of interest is the current through the circuit due to an applied voltage source. (Fig. a)
- For parallel resonance, the resonant curve of interest is the voltage across the output terminals of the network due to an applied current source. (Fig. b)

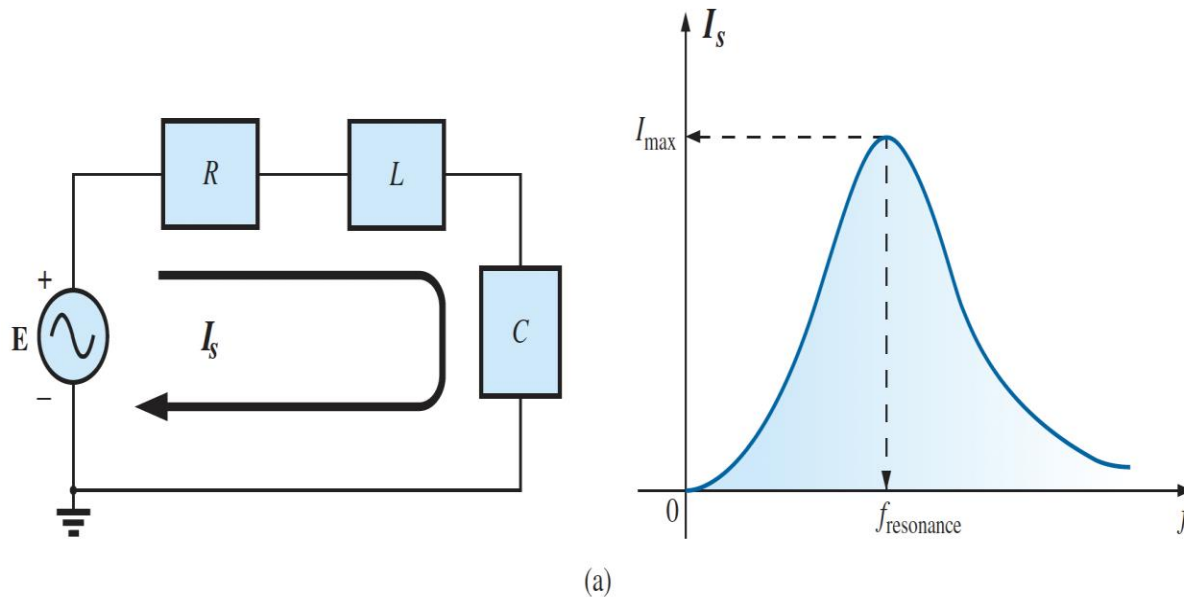


Fig. Series Resonant Circuit

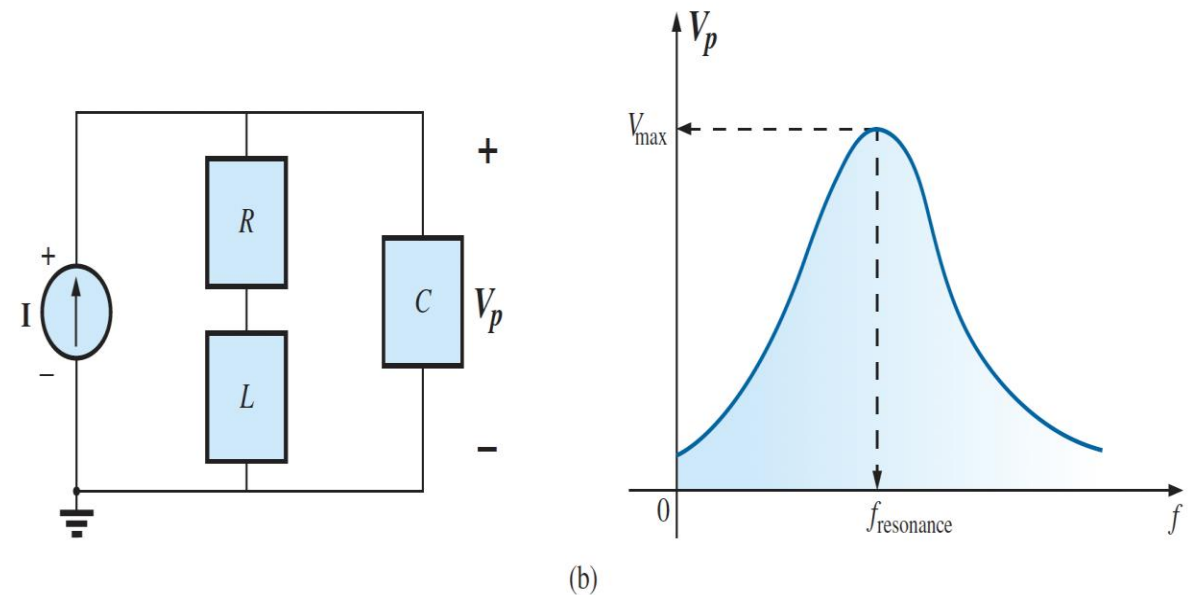


Fig. Parallel Resonant Circuit

Series Resonant Circuit

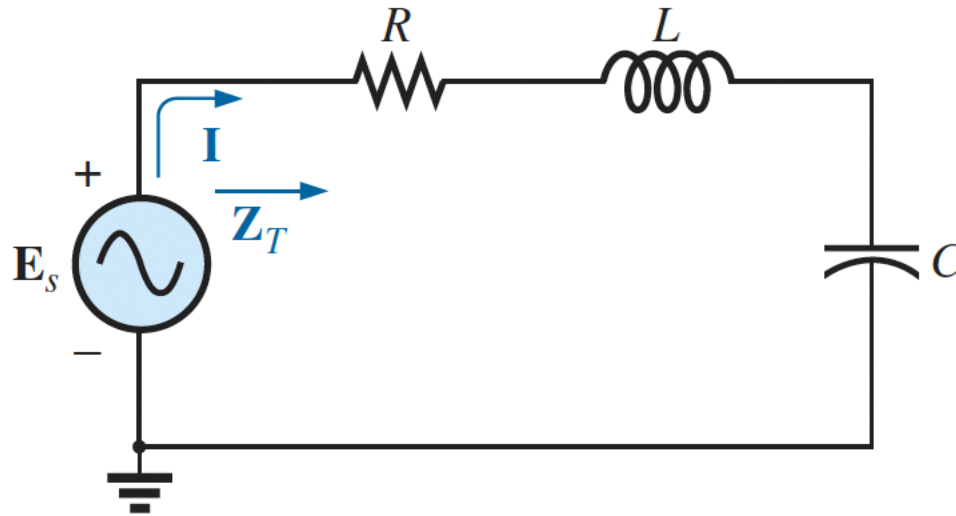


Fig. Series Resonant Circuit

Impedance vs Frequency

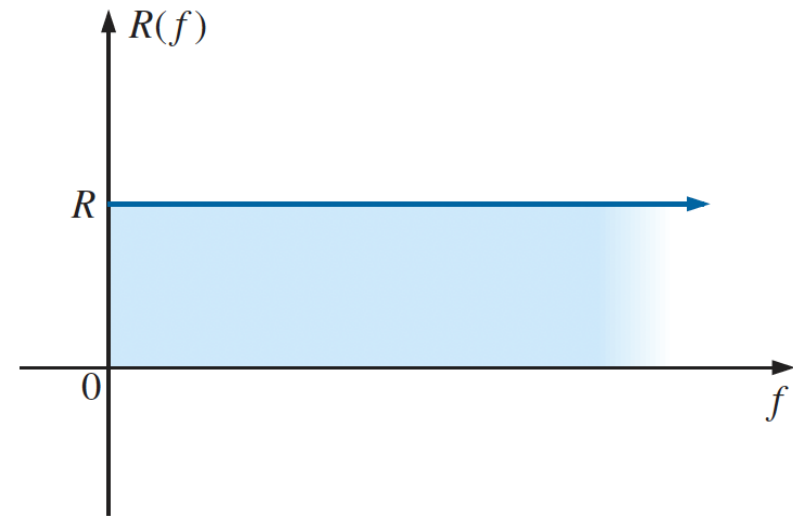


Fig. Resistance versus frequency

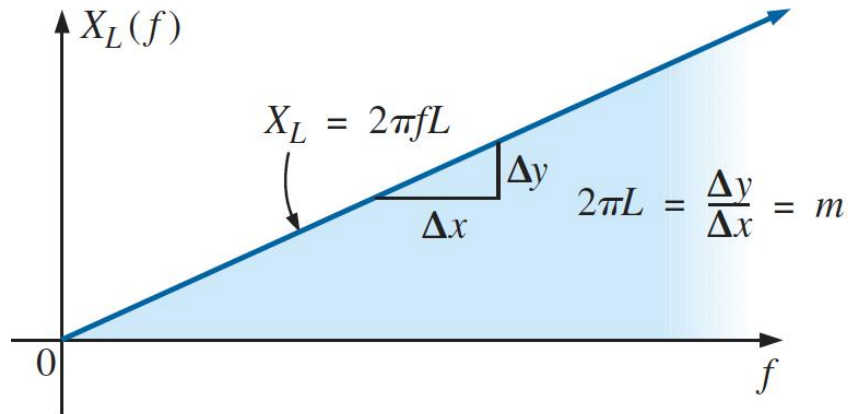


Fig. Inductive reactance versus frequency

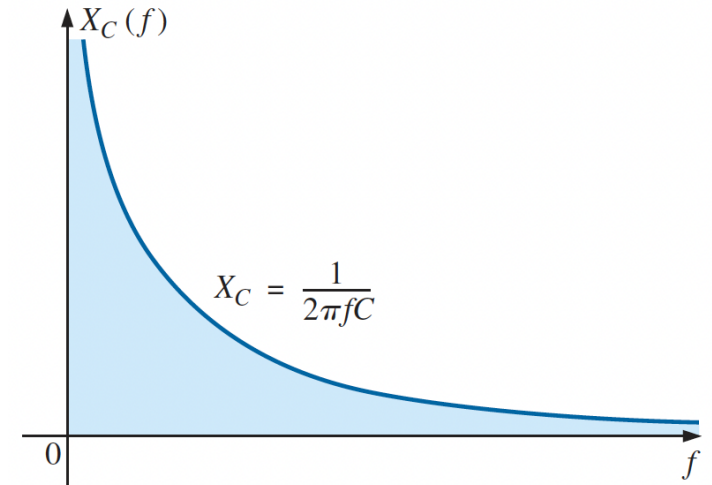


Fig. Capacitive reactance versus frequency

Impedance vs Frequency

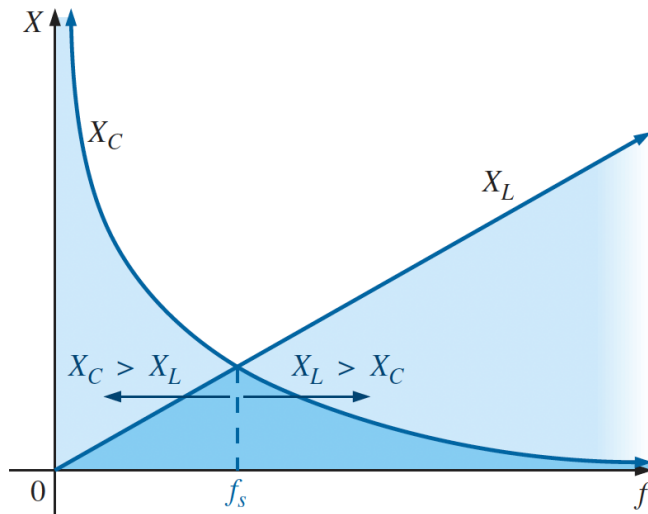


Fig. Placing the frequency response of the inductive and capacitive reactance of a series R-L-C circuit on the same set of axes

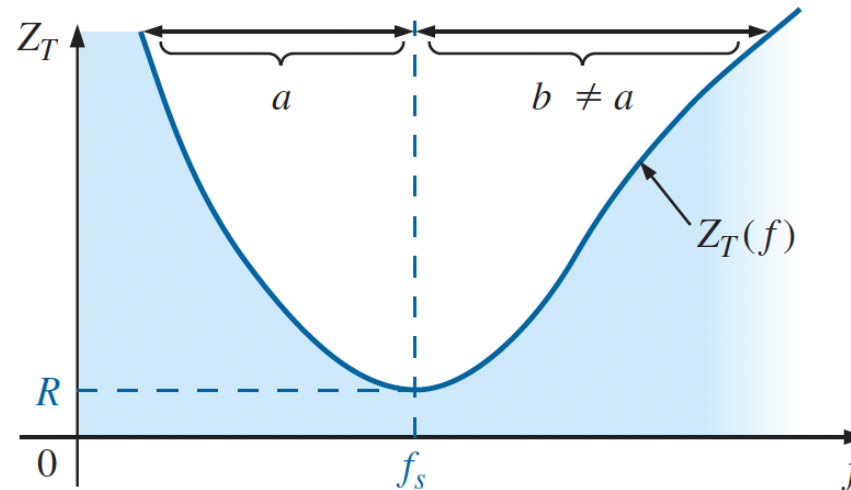


Fig. Z_T versus frequency for the series resonant circuit

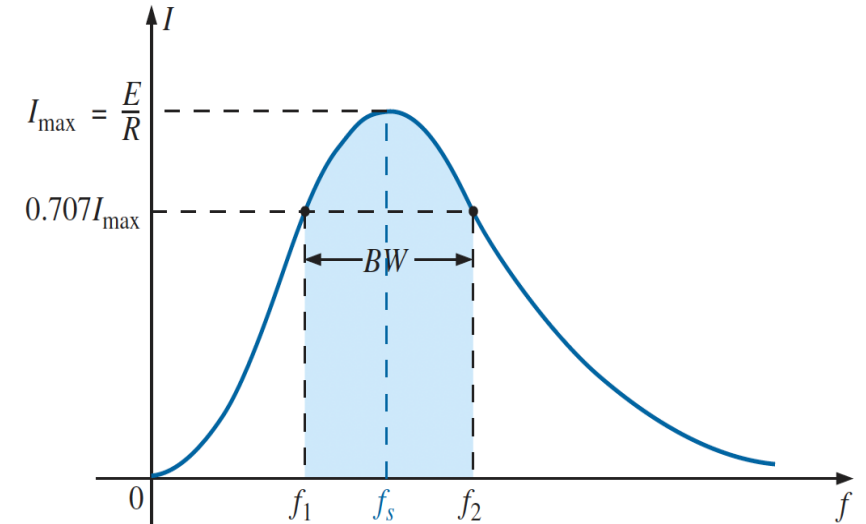


Fig. I versus frequency for the series resonant circuit

Series Resonant Circuit

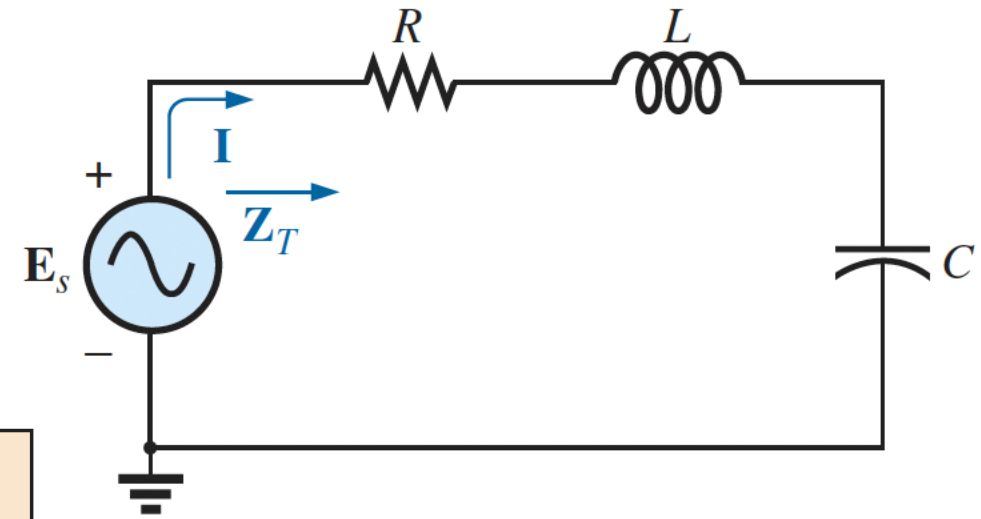


Fig. Series Resonant Circuit

Total
Impedance

$$Z_T = R + jX_L - jX_C = R + j(X_L - X_C)$$

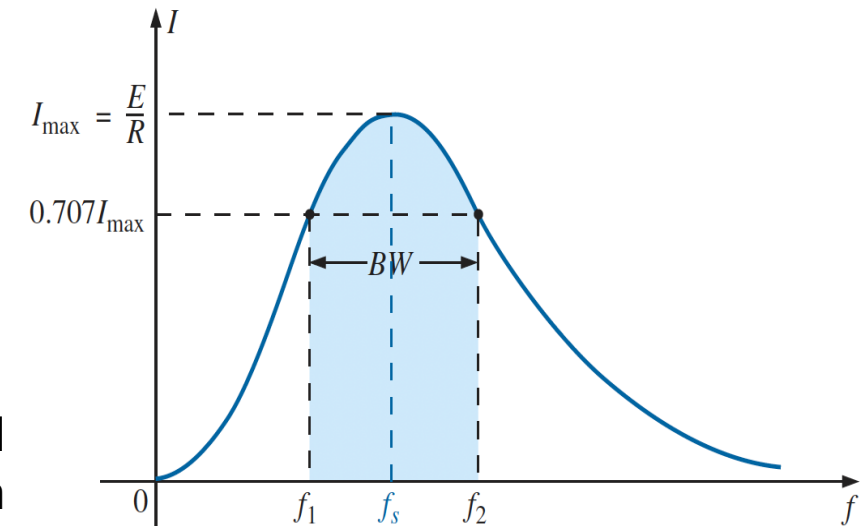
Resonance
Condition

$$X_L = X_C$$

Total
Impedance
at resonance

$$Z_{T_s} = R$$

It is the lowest possible value of the total impedance for the full frequency range resulting in the highest value for the current in the circuit.



*Fig. I versus frequency
for the series resonant
circuit*

Series Resonant Circuit

Resonant Frequency

$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

or

$$\omega_s = \frac{1}{\sqrt{LC}}$$

Peak Resonant Current

$$I_{\max} = \frac{E \angle 0^\circ}{R \angle 0^\circ} = \frac{E}{R} \angle 0^\circ$$

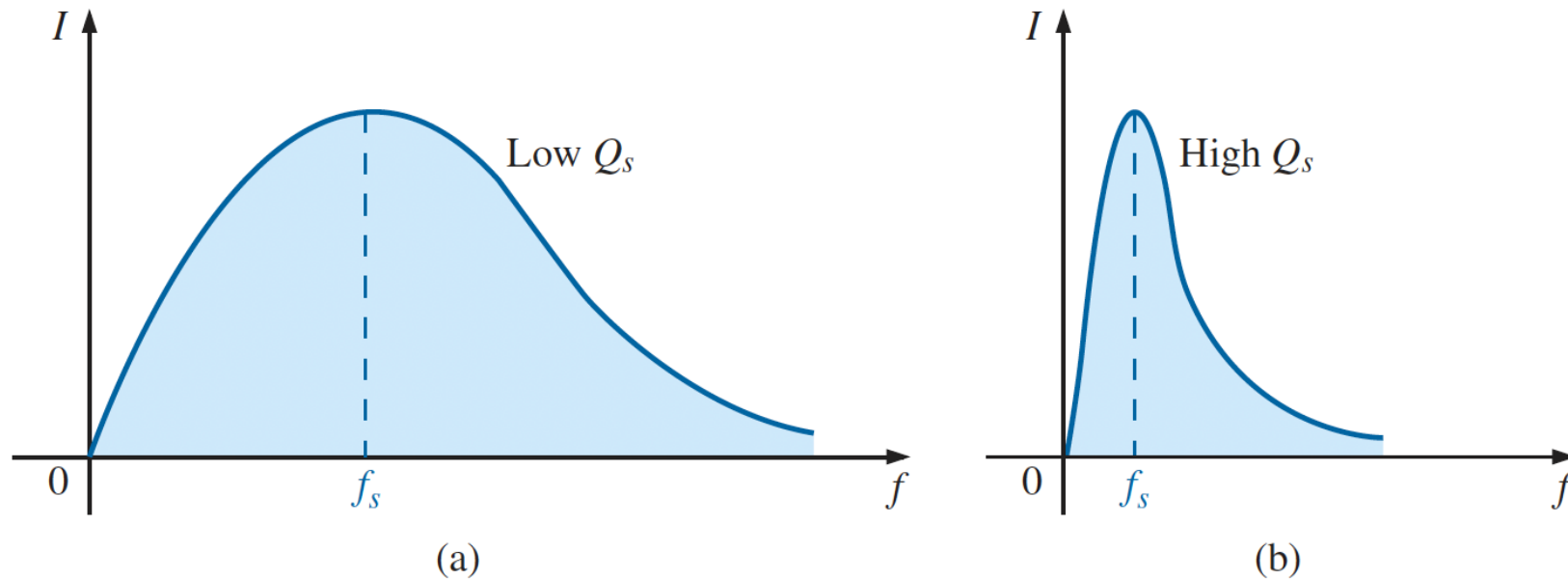
The input voltage and current are in phase at resonance.

Quality Factor (Q)

- The quality factor Q of a series resonant circuit is defined as the ratio of the reactive power of either the inductor or the capacitor to the average power of the resistor at resonance.
- The quality factor is also an indication of how much energy is placed in storage (continual transfer from one reactive element to the other) compared to that dissipated.
- The lower the level of dissipation for the same reactive power, the larger is the Qs factor and the more concentrated and intense is the region of resonance.

$$Q_s = \frac{\text{reactive power}}{\text{average power}}$$

Quality Factor (Q)



$$Q_s = \frac{\text{reactive power}}{\text{average power}}$$

Fig. Effect of Q_s as shape of resonant curve

Quality Factor (Q)

- Quality factor in terms of the circuit parameters:

$$Q_s = \frac{X_L}{R} = \frac{\omega_s L}{R}$$

$$Q_s = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Selectivity

- There is a **definite range of frequencies** at which the **current is near its maximum value** and the impedance is at a minimum. Those frequencies corresponding to **0.707 of the maximum current** are called the **band frequencies, cutoff frequencies, half-power frequencies, or corner frequencies**.
- They are indicated by f_1 and f_2 in the figure. The range of frequencies between the two is referred to as the bandwidth (abbreviated BW) of the resonant circuit.
- Since the resonant circuit is adjusted to “select” a band of frequencies, the curve in the figure is called the selectivity curve.

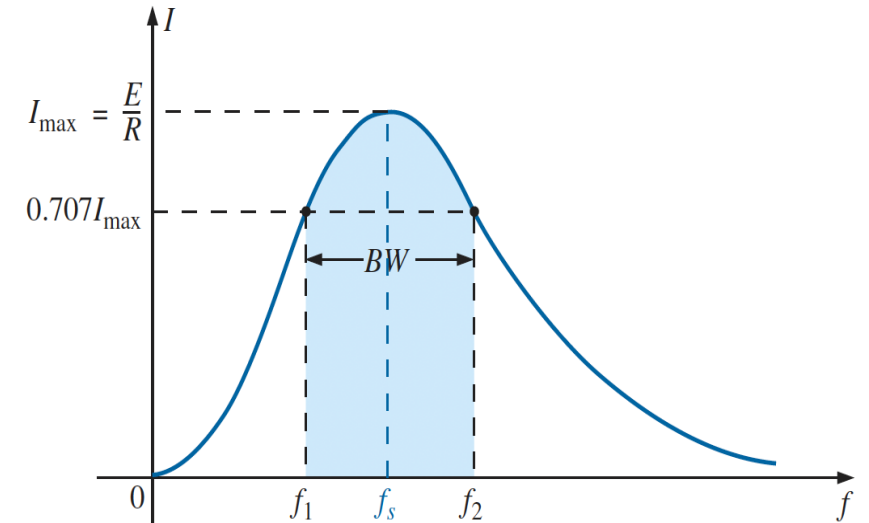


Fig. I versus frequency for the series resonant circuit

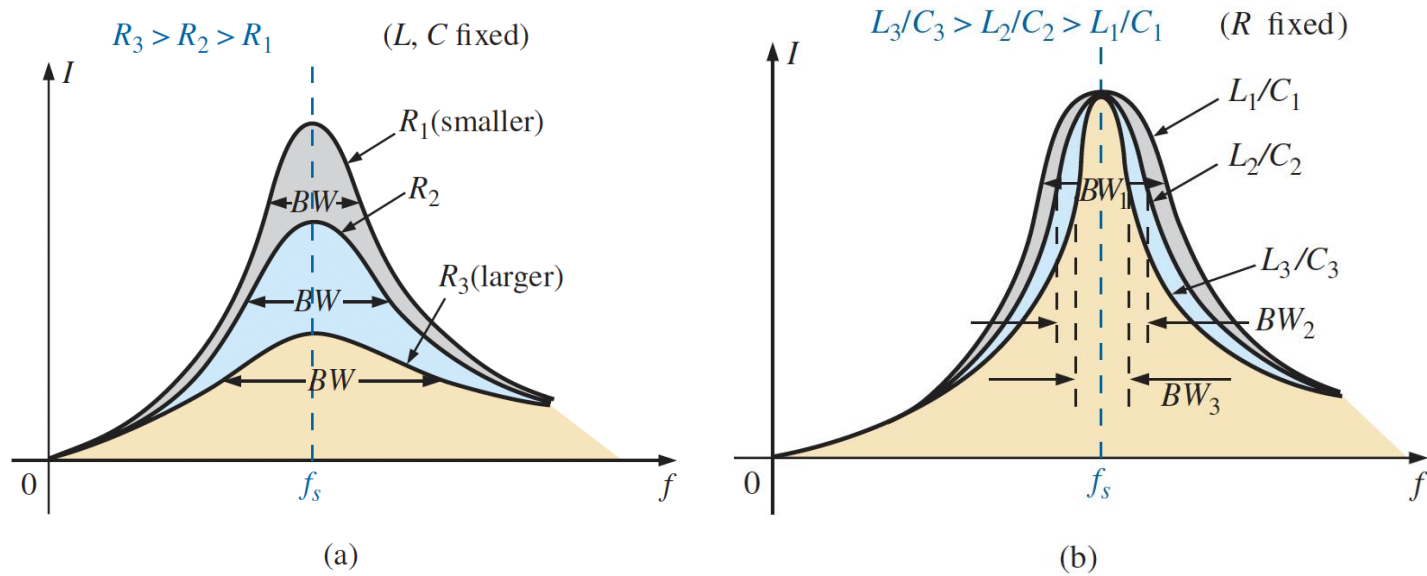
Selectivity

- Why are they called half-power frequencies?

Selectivity

- The smaller the bandwidth, the higher is the selectivity.

Selectivity



Effect of R , L , and C on the selectivity curve for the series resonant circuit.

Cutoff Frequencies

Cutoff Frequencies

Cutoff Frequencies

Lower
Cut-off frequency

$$f_1 = \frac{1}{2\pi} \left[-\frac{R}{2L} + \frac{1}{2} \sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}} \right] \text{ (Hz)}$$

Upper
Cut-off frequency

$$f_2 = \frac{1}{2\pi} \left[\frac{R}{2L} + \frac{1}{2} \sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}} \right] \text{ (Hz)}$$

Bandwidth

$$BW = f_2 - f_1 = \frac{R}{2\pi L}$$

Cutoff Frequencies

Cutoff Frequencies

Relation between
bandwidth and
quality factor

$$BW = \frac{f_s}{Q_s}$$

Larger the quality factor, smaller is the bandwidth, and vice versa

Fractional
bandwidth

$$\frac{f_2 - f_1}{f_s} = \frac{1}{Q_s}$$

An indication of the width of the bandwidth compared to the resonant frequency

Resonant
Frequency

$$f_s = \sqrt{f_1 f_2}$$

Series Resonant Circuit

Important equations related to series resonant circuits.

At Resonance

$$X_L = X_C$$

$$Z_{T_S} = R$$

$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

$$V_{L_S} = V_{C_S} = Q_s E$$

$$F_{P_S} = 1$$

$$Q_s = \frac{X_L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Additional Equations

$$P_{\text{HPF}} = \frac{1}{2}P_{\text{max}}$$

$$f_1 = \frac{1}{2\pi} \left[\frac{R}{2L} + \frac{1}{2} \sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}} \right]$$

$$f_2 = \frac{1}{2\pi} \left[-\frac{R}{2L} + \frac{1}{2} \sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}} \right]$$

$$BW = f_2 - f_1 = \frac{R}{2\pi L}$$

$$f_s = \sqrt{f_1 f_2}$$

Example #1

EXAMPLE 21.2 The bandwidth of a series resonant circuit is 400 Hz.

- a. If the resonant frequency is 4000 Hz, what is the value of Q_s ?
- b. If $R = 10\ \Omega$, what is the value of X_L at resonance?
- c. Find the inductance L and capacitance C of the circuit.

Example #2

EXAMPLE 21.4

- Determine the Q_s and bandwidth for the response curve in Fig. 21.24.
- For $C = 100 \text{ nF}$, determine L and R for the series resonant circuit.
- Determine the applied voltage.

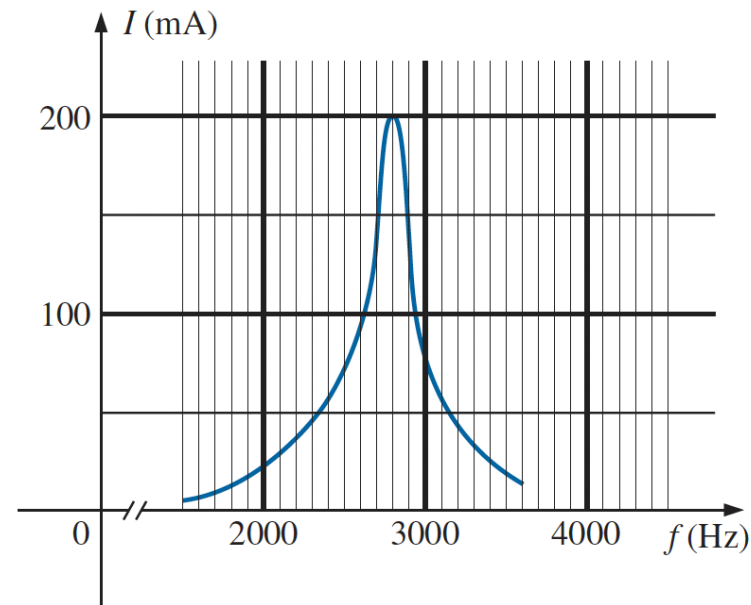


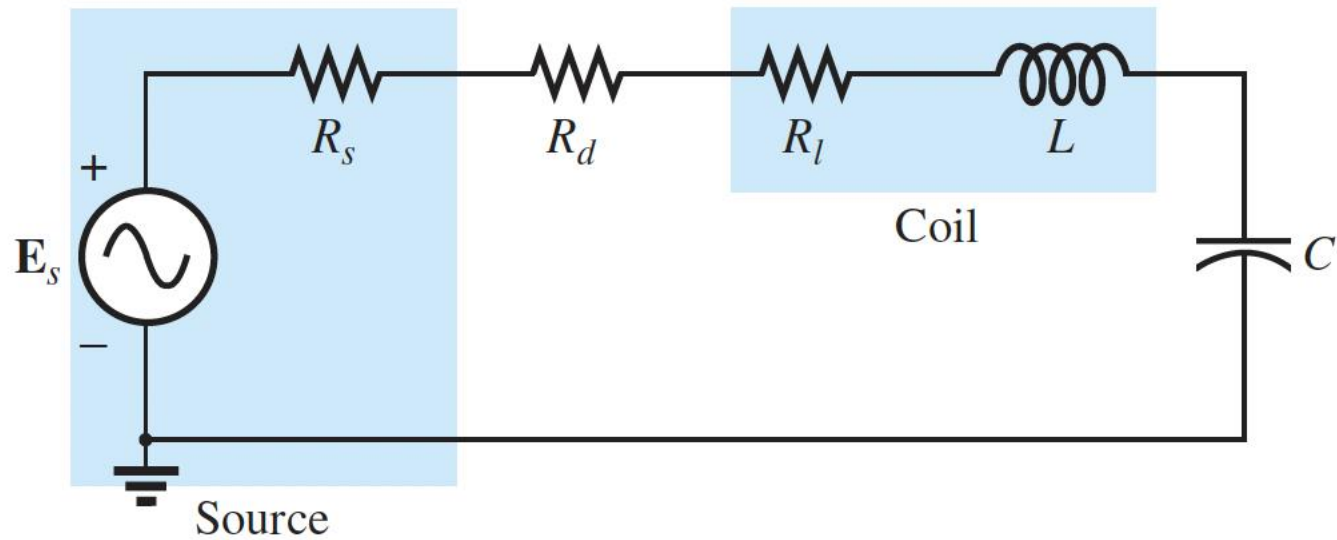
FIG. 21.24 Example #2

Example #3

EXAMPLE 21.5 A series R - L - C circuit is designed to resonate at $\omega_s = 10^5$ rad/s, have a bandwidth of $0.15\omega_s$, and draw 16 W from a 120 V source at resonance.

- a. Determine the value of R .
- b. Find the bandwidth in hertz.
- c. Find the nameplate values of L and C .
- d. Determine the Q_s of the circuit.
- e. Determine the fractional bandwidth.

Practical Considerations



$$R = R_s + R_d + R_l$$

Fig. Practical Series resonant circuit

Parallel Resonant Circuit

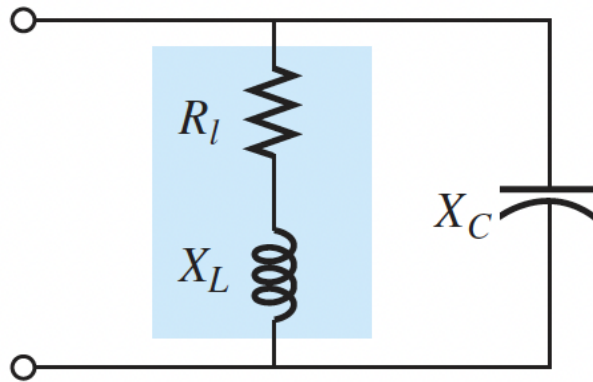


Fig. Practical parallel L-C network

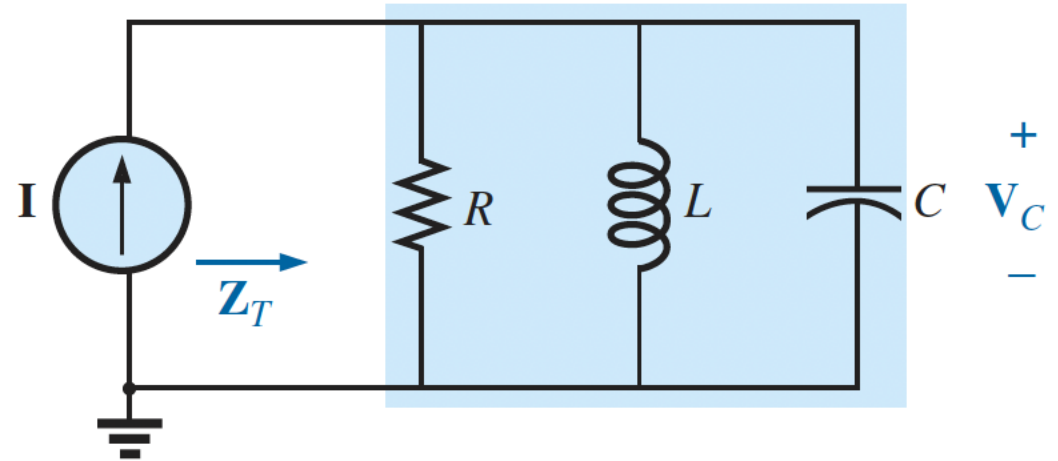


Fig. Ideal parallel resonant network

Parallel Resonant Circuit

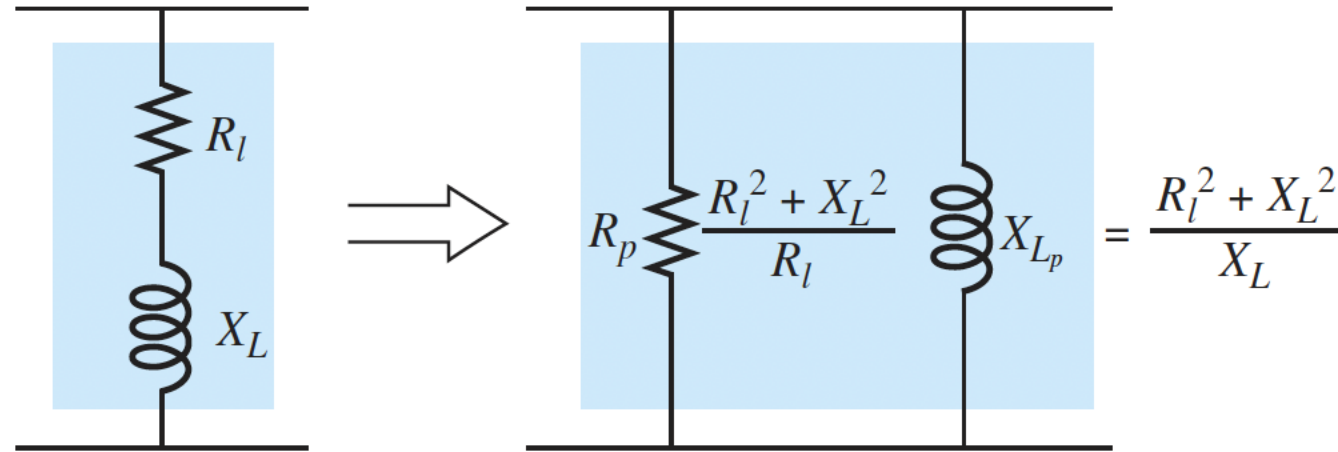


Fig. Equivalent parallel network for a series R-L combination

Parallel Resonant Circuit

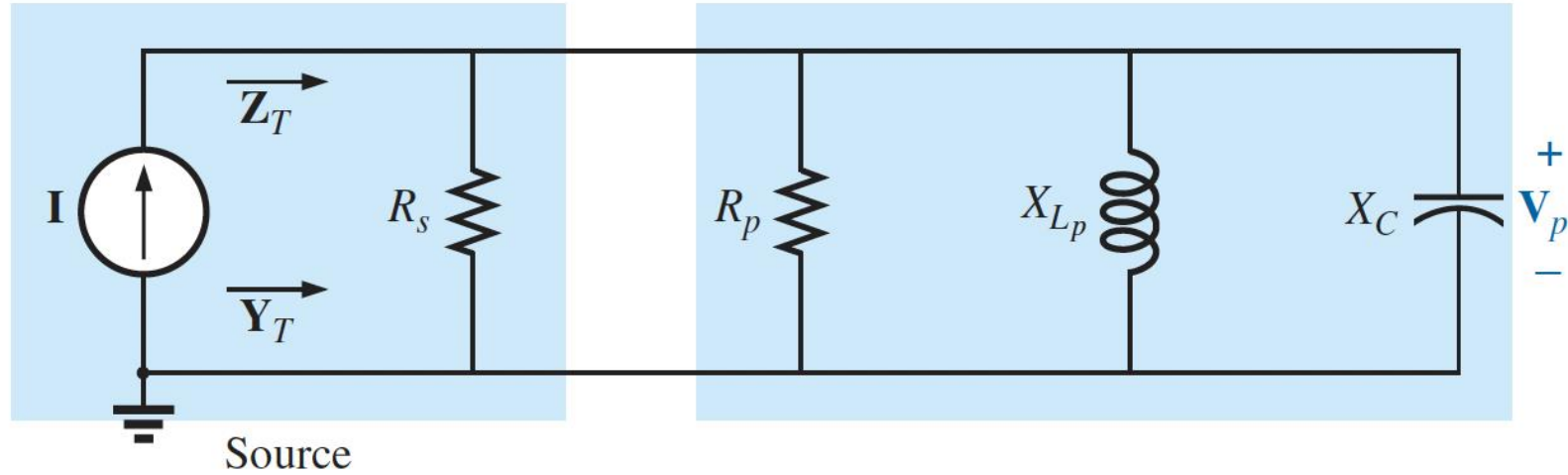


Fig. Substituting the equivalent parallel network for the series R-L combination

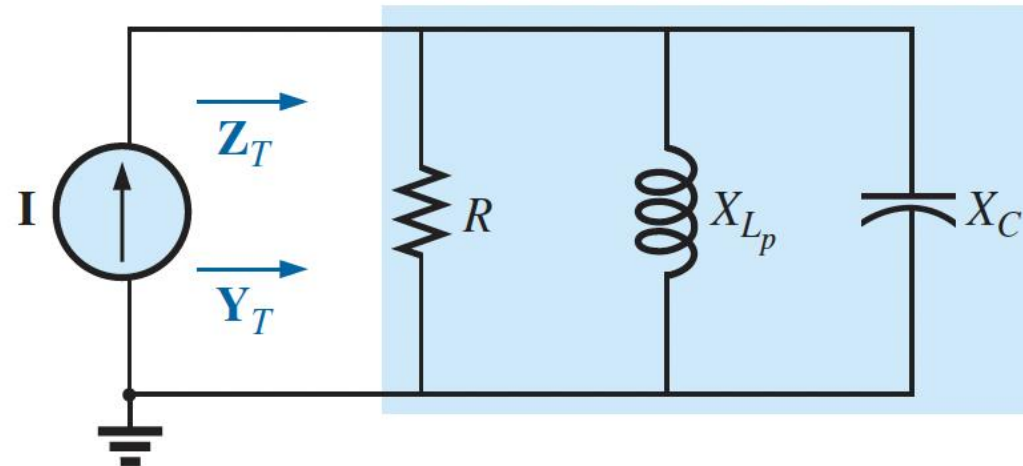


Fig. Substituting $R = R_s \parallel R_p$ for the network

Parallel Resonant Circuit

$$\mathbf{Y}_T = \frac{1}{R} + j\left(\frac{1}{X_C} - \frac{1}{X_{L_p}}\right)$$

For unity power factor, the reactive component must be zero.

$$X_{L_p} = X_C$$

$$f_p = f_s \sqrt{1 - \frac{R_l^2 C}{L}}$$

f_p is the resonant frequency of a parallel resonant circuit (for p.f.=1) and f_s is the resonant frequency as determined by $X_L = X_C$ for series resonance. Note that unlike a series resonant circuit, the resonant frequency f_p is a function of resistance (in this case R_l)

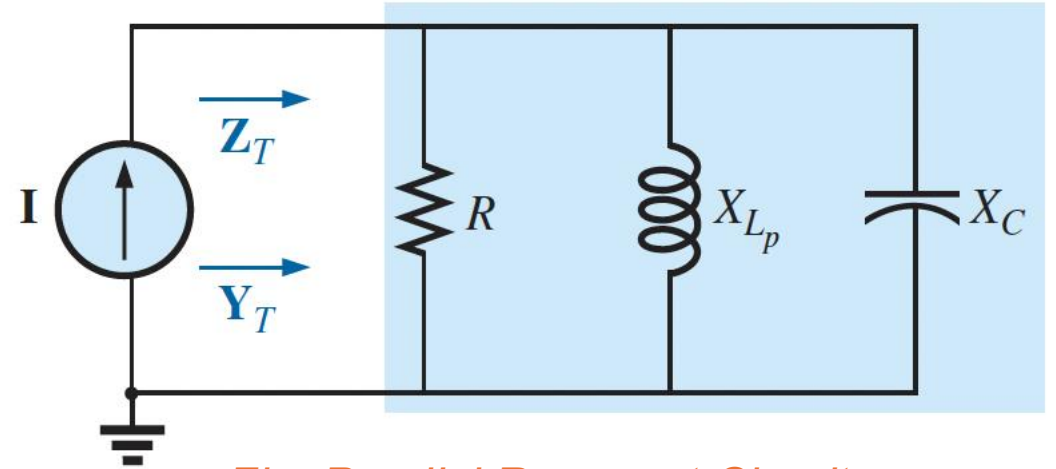


Fig. Parallel Resonant Circuit

Maximum Impedance Condition

The frequency at which **maximum impedance occurs** is defined by f_m and is slightly more than f_p .

$$f_m = f_s \sqrt{1 - \frac{1}{4} \left(\frac{R_l^2 C}{L} \right)}$$

$$f_s > f_m > f_p$$

$$Z_{T_m} = R \parallel X_{L_p} \parallel X_C \Big|_{f=f_m}$$

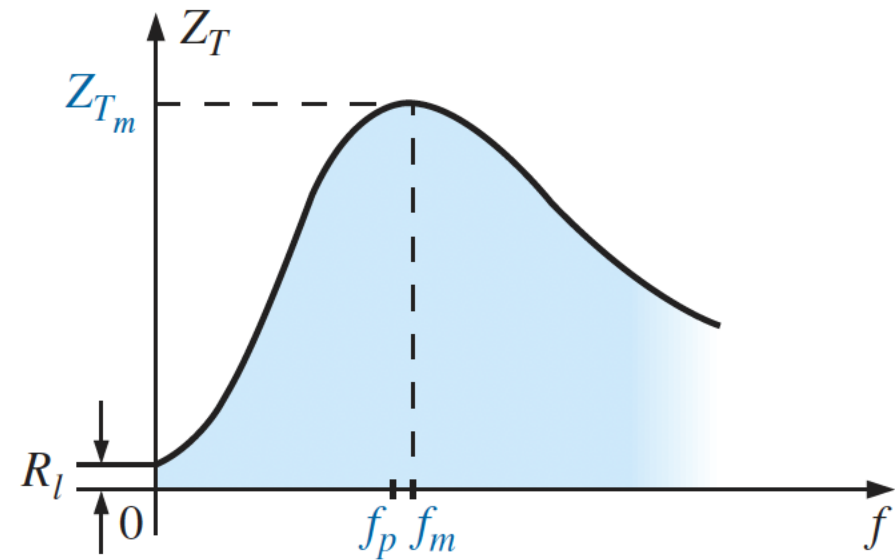


Fig. Z_T versus frequency for the parallel resonant circuit.

Quality Factor

$$Q_p = \frac{R}{X_{L_p}} = \frac{R_s \parallel R_p}{X_{L_p}}$$

$$Q_p = \frac{R_s \parallel R_p}{X_C}$$

$$Q_p = \frac{X_L}{R_1} = Q_l \quad R_s \gg R_p$$

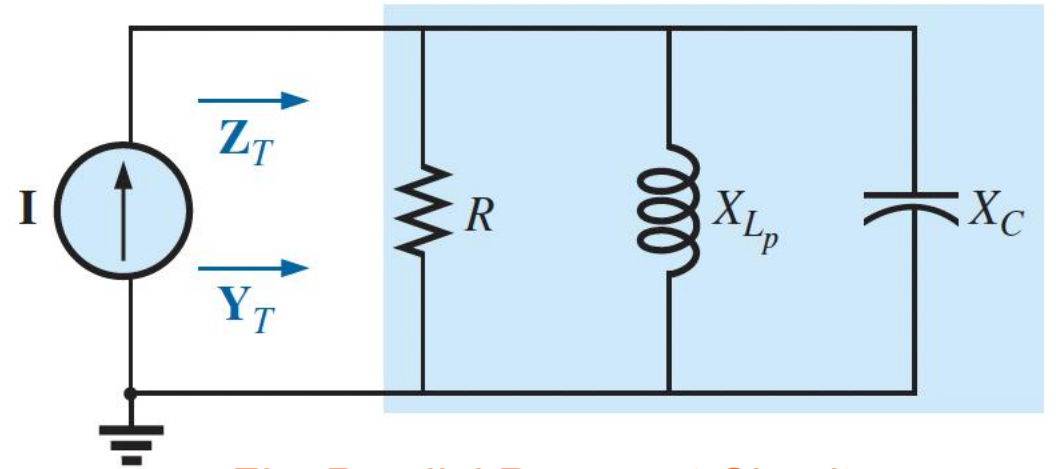


Fig. Parallel Resonant Circuit

Bandwidth

The half-power frequencies are defined by the condition that the output voltage is 0.707 times the maximum value.

$$f_1 = \frac{1}{4\pi C} \left[-\frac{1}{R} + \sqrt{\frac{1}{R^2} + \frac{4C}{L}} \right]$$

$$f_2 = \frac{1}{4\pi C} \left[\frac{1}{R} + \sqrt{\frac{1}{R^2} + \frac{4C}{L}} \right]$$

$$BW = f_2 - f_1 = \frac{f_p}{Q_p}$$

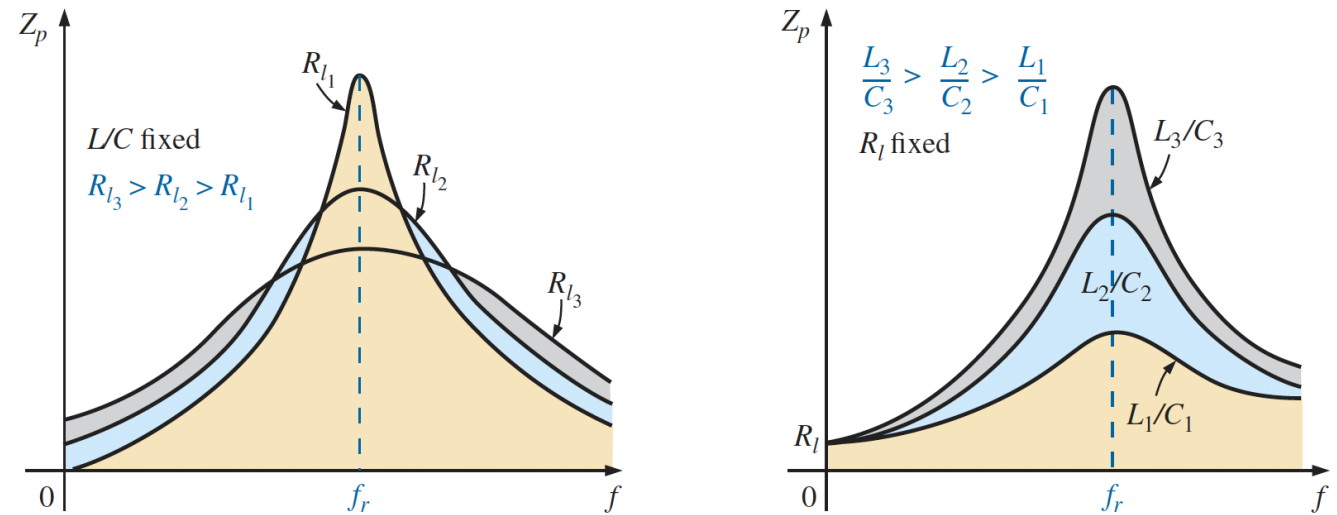


Fig. Effect of R_l , L , and C on the parallel resonance curve

Parallel Resonant Circuit

Parallel resonant circuit ($f_s = 1/(2\pi\sqrt{LC})$).

	Any Q_l	$Q_l \geq 10$	$Q_l \geq 10, R_s \gg Q_l^2 R_l$
f_p	$f_s \sqrt{1 - \frac{R_l^2 C}{L}}$	f_s	f_s
f_m	$f_s \sqrt{1 - \frac{1}{4} \left[\frac{R_l^2 C}{L} \right]}$	f_s	f_s
Z_{T_p}	$R_s \parallel R_p = R_s \left(\frac{R_l^2 + X_L^2}{R_l} \right)$	$R_s \parallel Q_l^2 R_l$	$Q_l^2 R_l$
Z_{T_m}	$R_s \parallel Z_{R-L} \parallel Z_C$	$R_s \parallel Q_l^2 R_l$	$Q_l^2 R_l$
Q_p	$\frac{Z_{T_p}}{X_{L_p}} = \frac{Z_{T_p}}{X_C}$	$\frac{Z_{T_p}}{X_L} = \frac{Z_{T_p}}{X_C}$	Q_l
BW	$\frac{f_p}{Q_p}$ or $\frac{f_m}{Q_p}$	$\frac{f_p}{Q_p} = \frac{f_s}{Q_p}$	$\frac{f_p}{Q_l} = \frac{f_s}{Q_l}$
I_L, I_C	Network analysis	$I_L = I_C = Q_l I_T$	$I_L = I_C = Q_l I_T$

Example #4

EXAMPLE 21.7 For the parallel resonant circuit in Fig. 21.37 with $R_s = \infty \Omega$:

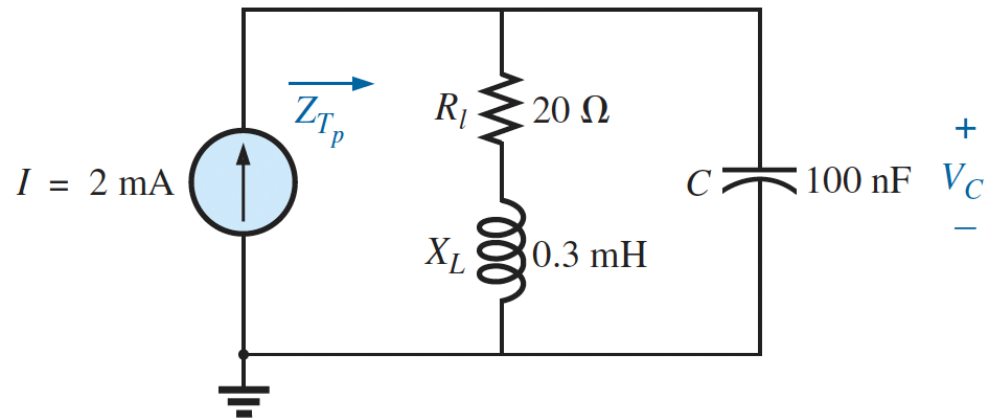


FIG. 21.37
Example 21.7.

- Determine f_s , f_m , and f_p , and compare their levels.
- Calculate the maximum impedance and the magnitude of the voltage V_C at f_m .
- Determine the quality factor Q_p .
- Calculate the bandwidth.

Example #5

EXAMPLE 21.8 For the network in Fig. 21.38 with f_p provided:

- Determine Q_l .
- Determine R_p .
- Calculate Z_{T_p} .
- Find C at resonance.
- Find Q_p .
- Calculate the BW and cutoff frequencies.

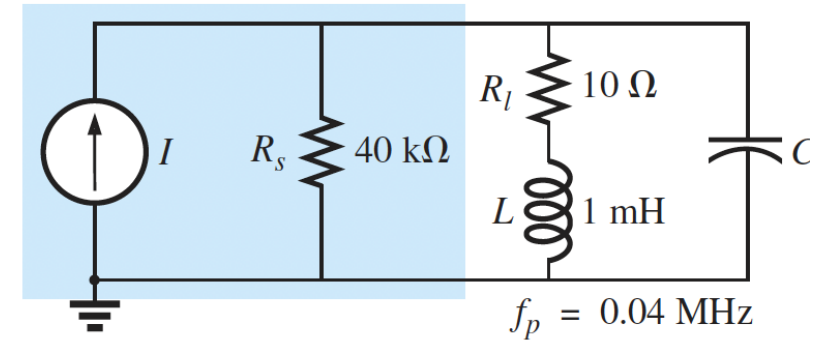


FIG. 21.38
Example 21.8.

Example #6

EXAMPLE 21.11 Design a parallel resonant circuit to have the response curve in Fig. 21.42 using a 1 mH, 10 Ω inductor and a current source with an internal resistance of 40 k Ω .

